

Pakistan Crop Knowledge Base: A Detailed Agricultural Reference

Chapter 1: Overview of Agriculture in Pakistan

1.1 Importance of Agriculture

Agriculture is the backbone of Pakistan's economy, contributing roughly 19-23 percent of national GDP and employing nearly 37-40 percent of the labor force. It supplies raw material to major industries such as textiles (cotton), sugar (sugarcane), and food processing, and it underpins food security for a population exceeding 240 million people. The sector is dominated by smallholder farmers, with the majority of farms under 12.5 acres (5 hectares), meaning that advisory and extension services aimed at small farm efficiency have outsized national impact.

1.2 Agro-Ecological Zones of Pakistan

Pakistan is divided into distinct agro-ecological zones based on climate, soil, and water availability:

- Indus Delta Zone (southern Sindh): saline soils, rice, banana, and fish farming.
- Southern Irrigated Plain (Sindh): cotton, wheat, rice, sugarcane, banana, mango.
- Northern Irrigated Plain (Punjab): the country's breadbasket; wheat, cotton, rice, sugarcane, maize, citrus.
- Barani (rain-fed) Lands (Punjab Potohar region, parts of KP): wheat, groundnut, pulses, low-input farming dependent on monsoon rains.
- Wet Mountains (KP, Azad Kashmir, Gilgit-Baltistan): maize, apple, apricot, walnut, terraced farming.
- Northern Dry Mountains (Gilgit-Baltistan, Chitral): apple, apricot, cherry, potato, short growing season.
- Sandy Desert (Cholisthan, Thar): livestock-dominant, guar, bajra (pearl millet), drought-tolerant crops.
- Western Dry Mountains (Balochistan): apple, grapes, cherry, almond, deciduous fruit orchards, dryland wheat.
- Dry Western Plateau (Balochistan): wheat, sorghum, fruit orchards, groundwater-dependent.

1.3 Major Cropping Seasons

Pakistani agriculture operates on two principal cropping seasons:

- Kharif (summer season): sowing April-July, harvesting September-December. Key crops: rice, cotton, sugarcane, maize (spring and autumn maize also exist), moong, mash, sorghum, bajra, groundnut, chili, okra, cucurbits.

- Rabi (winter season): sowing October-December, harvesting March-May. Key crops: wheat, gram (chickpea), lentil (masoor), mustard/canola, barley, potato (autumn crop also grown), peas, onion, tomato (in some zones), berseem fodder.

1.4 Soil Types Found in Pakistan

- Alluvial soils: found along the Indus River plains (Punjab, Sindh); highly fertile, well suited to wheat, rice, sugarcane, cotton.
- Loess/desert soils: Potohar and Thal areas; low organic matter, moderate fertility, require careful moisture management.
- Saline and sodic soils: widespread in lower Sindh and parts of southern Punjab due to poor drainage and high water tables; require gypsum application and leaching for reclamation.
- Mountain soils: shallow, stony, found in KP, GB, and AJK; suited to orchards and terraced cultivation.
- Sandy desert soils: Cholistan and Thar; low water-holding capacity, suited to drought-tolerant crops and livestock grazing.

1.5 Irrigation Systems

Pakistan relies on the Indus Basin Irrigation System (IBIS), the largest contiguous irrigation network in the world, fed by the Indus, Jhelum, Chenab, Ravi, and Sutlej rivers and regulated through barrages (e.g., Sukkur, Guddu, Taunsa, Kotri) and large reservoirs (Tarbela, Mangla, Diamer-Bhasha under construction). Key irrigation methods used by farmers include:

- Flood/basin irrigation (traditional, most common, water-inefficient).
- Furrow irrigation (used for row crops like maize, cotton, sugarcane).
- Tubewell irrigation (groundwater extraction, common in Punjab where canal supply is insufficient; over-extraction is causing water table decline in some districts and salinity in others).
- Drip and sprinkler irrigation (increasingly promoted for high-value crops such as orchards, vegetables, and cotton under government subsidy schemes such as the Punjab Drip Irrigation Programme).
- Watercourse lining programs aim to reduce the roughly 25-30 percent conveyance losses typical in unlined earthen watercourses.

1.6 General Fertilizer Guidance

Standard fertilizer nutrients referenced throughout this document are Nitrogen (N), Phosphorus (P₂O₅), and Potassium (K₂O), typically applied through Urea (46% N), Diammonium Phosphate/DAP (18% N, 46% P₂O₅), Single Super Phosphate/SSP (16% P₂O₅), and Sulphate of Potash/SOP or Muriate of Potash/MOP (60% K₂O). Nitrogen is usually split-applied (part at sowing, remainder at later growth stages) because it is prone to leaching and volatilization, while phosphorus and potassium are typically applied fully at sowing since they are less mobile in soil. Micronutrient deficiencies of zinc (common in rice and wheat on calcareous soils), boron (cotton, fruit crops), and iron (citrus, in calcareous/alkaline

soils causing chlorosis) are widespread across Pakistani soils and are corrected with foliar sprays or soil-applied zinc sulphate, borax, and iron chelates respectively.

1.7 Integrated Pest Management (IPM) Principles

Across all crops discussed in this reference, the following general IPM principles apply and are recommended by Pakistan's provincial agriculture extension departments:

- Use certified, disease-free seed and resistant/tolerant varieties where available.
- Practice crop rotation to break pest and disease cycles (e.g., avoiding continuous rice-wheat or cotton monocropping on the same field).
- Monitor fields regularly (scouting) to detect pest thresholds before applying chemical control.
- Conserve natural enemies (ladybird beetles, parasitic wasps, spiders) by minimizing broad-spectrum insecticide use.
- Use pheromone traps and light traps for early pest detection (especially for cotton bollworm, fruit fly, and armyworm).
- Apply pesticides only when economic thresholds are reached, rotate chemical classes to prevent resistance, and observe pre-harvest intervals (PHI) for food safety.
- Remove and destroy crop residues and volunteer plants that harbor pests and inoculum between seasons.

1.8 Climate Change Considerations

Pakistan is highly vulnerable to climate variability. Rising temperatures are shortening the grain-filling period of wheat and rice, increasing heat stress during flowering in cotton and mango, and shifting pest distributions northward (e.g., fall armyworm spread in maize since 2019, increased whitefly pressure in cotton). Erratic monsoon patterns have caused both severe flooding (notably 2010 and 2022) and prolonged droughts (notably in Sindh and Balochistan), damaging standing crops and infrastructure. Adaptation strategies promoted by agricultural research institutes (e.g., National Agricultural Research Centre - NARC, Ayub Agricultural Research Institute - AARI) include heat- and drought-tolerant variety development, laser land leveling for water conservation, and shifting sowing windows.

1.9 Government Support and Extension Structure

Agricultural extension in Pakistan is a provincial subject following the 18th Constitutional Amendment. Key institutions include:

- Provincial Agriculture Departments (Punjab, Sindh, KP, Balochistan) with district-level extension offices.
- Pakistan Agricultural Research Council (PARC) at the federal level, coordinating national research.
- Provincial research institutes: Ayub Agricultural Research Institute (AARI, Faisalabad), Sindh Agriculture University Tandojam, Agricultural Research Institute Tarnab (KP), Arid Zone Research

Institute (Balochistan).

- Kissan Card and agricultural credit schemes offering subsidized loans for inputs.
- Zarai Taraqiati Bank Limited (ZTBL), the main agricultural development bank providing crop loans.
- Provincial seed corporations and the Federal Seed Certification and Registration Department (FSC&RD;) responsible for seed quality certification.

Chapter 2: Wheat (*Triticum aestivum*) — Gandum

2.1 Overview and Importance

Wheat is Pakistan's single most important staple crop and the largest contributor to national cereal production, occupying roughly 8.5-9.0 million hectares annually with total production typically in the range of 25-28 million tonnes. It is grown in every province, with Punjab contributing the largest share (about 70-75 percent of national output), followed by Sindh, Khyber Pakhtunkhwa, and Balochistan. Wheat flour (atta) is the primary source of calories for most Pakistani households, and the government sets a minimum support price (MSP) each year that heavily influences farmer planting decisions and food inflation.

2.2 Growing Regions

- Punjab: Faisalabad, Sahiwal, Multan, Bahawalpur, Rahim Yar Khan, Vehari, Sargodha — irrigated plains producing the highest yields.
- Sindh: Sukkur, Larkana, Nawabshah, Hyderabad, Thatta.
- Khyber Pakhtunkhwa: Peshawar valley, Mardan, Swabi, Dera Ismail Khan.
- Balochistan: Quetta, Pishin, Loralai — largely rain-fed/dryland wheat with lower yields.
- Barani (rain-fed) tracts: Potohar Plateau (Rawalpindi, Chakwal, Jhelum, Attock) depend entirely on winter rainfall.

2.3 Climate and Soil Requirements

Wheat requires a cool growing season with temperatures between 10-25°C during vegetative growth and slightly warmer, dry conditions (around 25-30°C) during grain-filling and maturity for good grain quality. Frost at flowering can cause sterility, while excessive heat above 32-35°C during grain fill (a growing problem with late-sown wheat and climate warming) reduces grain weight ("terminal heat stress"). Wheat performs best on well-drained loam to clay-loam alluvial soils with pH 6.0-7.5, though it tolerates a wider soil range better than rice or sugarcane. Waterlogged or highly saline soils reduce germination and yield significantly.

2.4 Major Varieties Cultivated in Pakistan

- Punjab: Faisalabad-2008, Ujala-2016, Ankor, Zincol-2016 (biofortified with zinc), Anaj-2017, Akbar-2019, MH-2020, Subhani-2021, Dilkash-2020.
- Sindh: TD-1, Sindh-81, Kiran-95, TJ-83, Sarsabz, Sassui, Benazir-2013.
- KP: Pirsabak-2013, Pirsabak-2015, Khyber-87, Fakhr-e-Sarhad.
- Zinc-biofortified varieties (Zincol-2016, Akbar-2019) are being promoted nationally by HarvestPlus/PARC to combat zinc deficiency in the population.

- Rust-resistant varieties are prioritized because Pakistan sits on a major wheat rust migration corridor (yellow/stripe rust, *Puccinia striiformis*, is the most economically damaging disease historically, along with leaf and stem rust).

2.5 Land Preparation

Standard practice is 2-3 ploughings followed by planking (soil leveling) to ensure uniform irrigation. Laser land leveling is increasingly promoted to improve water use efficiency, reducing irrigation water requirement by up to 25-30 percent and improving yield uniformity. In rice-wheat rotation areas of central Punjab, zero tillage (direct drilling wheat into rice stubble) and happy seeder technology are promoted to reduce cost, conserve moisture, and avoid stubble burning, which is a major contributor to winter smog in Punjab.

2.6 Sowing Time, Method, and Seed Rate

Optimum sowing window is 1 November to 15 December for irrigated areas of Punjab and Sindh (early-to-mid November is considered ideal for maximum yield potential; sowing after mid-December sharply reduces yield due to shortened growth duration and terminal heat stress). Barani (rain-fed) areas typically sow slightly earlier, from mid-October, to capture available soil moisture. Seed rate is generally 100-125 kg/ha (45-50 kg per acre) for timely sowing, increased to 150 kg/ha for late sowing to compensate for reduced tillering. Row spacing is typically 22.5 cm using seed-cum-fertilizer drills, which is preferred over broadcasting for better input use efficiency.

2.7 Irrigation Schedule

Wheat typically requires 4-6 irrigations depending on rainfall and soil type:

- First irrigation (Chillawa/CRI - Crown Root Initiation stage): 20-25 days after sowing — considered the most critical irrigation.
- Second irrigation: at tillering stage (around 40-45 days after sowing).
- Third irrigation: at jointing stage (60-65 days).
- Fourth irrigation: at flowering/booting stage (80-85 days) — second most critical for yield.
- Fifth irrigation: at milk/dough stage of grain filling (100-105 days).

Water stress at CRI and flowering stages causes the largest yield penalties.

2.8 Fertilizer/Nutrient Management

Recommended general dose for irrigated wheat is approximately 120-150 kg N, 85-115 kg P₂O₅, and 60 kg K₂O per hectare (roughly 50-60 kg N, 35-45 kg P₂O₅, 25 kg K₂O per acre), adjusted by soil test where available. Full phosphorus, potassium, and one-third to half of nitrogen are applied at sowing (basal), with the remaining nitrogen split between the first and second irrigations. Zinc sulphate (25 kg/ha) is commonly recommended as a basal application on zinc-deficient calcareous soils, particularly in rice-wheat systems.

2.9 Weed Management

Major weeds include *Phalaris minor* (canary grass, the most problematic and increasingly herbicide-resistant weed in Punjab's rice-wheat belt), wild oats (*Avena fatua*), *Chenopodium album* (bathu), and *Convolvulus arvensis*. Control combines pre-emergence herbicides (pendimethalin), post-emergence graminicides for grassy weeds, and broadleaf herbicides (2,4-D) applied at the tillering stage. Herbicide resistance in *Phalaris minor* has become a serious regional problem requiring herbicide rotation.

2.10 Major Pests

- Aphids (*Rhopalosiphum* spp., *Sitobion avenae*): infest leaves and ears, transmit barley yellow dwarf virus; controlled with imidacloprid or conservation of natural predators (ladybird beetles) below threshold levels.
- Termites: damage roots and stems, especially in light/sandy soils and rain-fed areas; controlled with seed treatment (chlorpyrifos-based) and field sanitation.
- Armyworm (occasional outbreaks): defoliates young plants; controlled with recommended insecticides when threshold is exceeded.

2.11 Major Diseases

- Yellow/stripe rust (*Puccinia striiformis*): the most economically important disease; yellow stripes of pustules on leaves; managed through resistant varieties, fungicide (propiconazole/tebuconazole) sprays, and early sowing to avoid peak disease pressure.
- Leaf rust and stem rust: similar management via resistant varieties and fungicides.
- Karnal bunt (*Tilletia indica*): affects grain quality and export marketability; managed via resistant varieties and seed treatment.
- Loose smut (*Ustilago tritici*): seed-borne; controlled with certified seed and systemic fungicide seed treatment (carboxin, tebuconazole).
- Powdery mildew: occurs in humid, cool conditions; managed with sulfur-based fungicides.

2.12 Harvesting and Yield

Wheat matures in 140-160 days from sowing depending on variety and sowing date, typically harvested late March to early May depending on region (Sindh harvests earliest, Punjab and KP later, Balochistan and high-altitude areas latest). Combine harvesters are now widely used in Punjab and Sindh, while manual harvesting with sickles persists in smaller and hilly farms. National average yield is around 2.9-3.2 tonnes/ha, though progressive irrigated farms using certified seed and full recommended input packages can achieve 4.5-5.5 tonnes/ha. Straw is retained as important livestock fodder (bhoosa/tor).

2.13 Post-Harvest and Storage

Grain moisture should be reduced to below 12 percent before storage to prevent fungal growth (aflatoxin risk) and insect infestation. Traditional storage uses jute bags in godowns; PASSCO (Pakistan Agricultural Storage and Services Corporation) and provincial food departments procure and store wheat in bag and bulk silo storage for the strategic reserve. Major storage pests include khapra beetle (*Trogoderma granarium*, a quarantine pest of major export concern) and rice weevil (*Sitophilus oryzae*), managed through fumigation (aluminum phosphide) and hermetic storage bags (e.g., PICS bags) at the smallholder level.

2.14 Economics and Marketing

The government announces a wheat support price before each Rabi season, which anchors farm-gate prices; the Punjab and Sindh food departments and PASSCO procure wheat directly from farmers at this rate to build the strategic reserve, though implementation and procurement targets vary by year and province. Wheat husk (bran/choker) is a significant by-product used in the poultry and livestock feed industry.

Chapter 3: Rice (*Oryza sativa*) — Chawal

3.1 Overview and Importance

Rice is Pakistan's second major staple cereal and its largest agricultural export earner, particularly the aromatic long-grain Basmati rice grown in the Kalar tract of central Punjab, which commands premium prices in Middle Eastern and European markets. Pakistan is among the top rice-exporting countries in the world. Total area under rice is roughly 2.7-3.0 million hectares with production around 8-9 million tonnes (paddy).

3.2 Growing Regions

- Punjab (Basmati belt): Gujranwala, Sheikhpura, Hafizabad, Sialkot, Narowal, Lahore — famous for Basmati rice grown under the traditional "Kalar tract" conditions.
- Punjab (coarse/IRRI rice): Sahiwal and southern districts grow coarse varieties for domestic consumption and export to price-sensitive markets.
- Sindh: Larkana, Shikarpur, Dadu, Thatta — major coarse rice production, including the famous Sindhri and IRRI varieties, and some Basmati in northern Sindh.
- Balochistan and KP: smaller pockets of cultivation, primarily for local consumption.

3.3 Climate and Soil Requirements

Rice is a semi-aquatic crop requiring standing water for most of its growth cycle, and thrives in hot, humid conditions with temperatures of 25-35°C. It performs well on heavy clay and clay-loam soils with poor drainage (which is normally a disadvantage for other crops but ideal for rice because it retains standing water). Basmati rice specifically requires the distinctive climate and soil of the Kalar tract to develop its characteristic aroma and elongation-on-cooking quality; attempts to grow Basmati outside this belt typically produce inferior grain quality.

3.4 Major Varieties Cultivated in Pakistan

- Basmati: Basmati-385 (traditional, long duration, prized aroma), Super Basmati (higher yielding, widely grown), Basmati-2000, Kissan Basmati, Shaheen Basmati, and newer high-yielding aromatic lines such as Basmati-515 and PB-1 (Punjab Basmati-1) developed for higher yield while retaining export quality traits.
- Coarse/IRRI type: IRRI-6, IRRI-9, KSK-133, KSK-434 — higher yielding, shorter duration, used for domestic bulk consumption and export to African and other price-sensitive markets.
- Sindh varieties: Sindhri Basmati (Basmati grown in northern Sindh), Shua-92, DR-83, NIA varieties from Nuclear Institute of Agriculture Tandojam.

3.5 Nursery Raising and Transplanting

Rice is grown via a two-stage process: nursery raising followed by transplanting (the traditional and still dominant method), or direct seeding (increasingly promoted to save water and labor). Nursery seed rate is about 20-25 kg/ha of seed sown on raised, puddled nursery beds. Seedlings are transplanted 25-35 days after sowing into puddled, leveled main fields, typically at a spacing of 22.5 x 15 cm (2-3 seedlings per hill), giving a plant population appropriate for full canopy development. Transplanting window for Basmati is typically mid-July to early August in Punjab; for coarse rice it can start slightly earlier (June).

3.6 Direct Seeded Rice (DSR) and Water-Saving Techniques

Because traditional puddled-transplanted rice consumes extremely high water volumes (estimated 4,000-5,000 liters of water per kg of rice produced under flooded conditions), water-saving alternatives are being promoted:

- Direct Seeded Rice (DSR): seed drilled directly into prepared, non-puddled fields, saving significant labor and water, though weed management is more challenging.
- Alternate Wetting and Drying (AWD): fields allowed to dry to a set threshold before re-flooding, reducing water use by 15-30 percent with minimal yield penalty.
- Laser leveling: improves water distribution uniformity and reduces losses.

3.7 Fertilizer/Nutrient Management

Recommended dose for Basmati rice is around 80-100 kg N, 60 kg P₂O₅, and 60 kg K₂O per hectare (roughly 35-45 kg N, 25 kg P₂O₅, 25 kg K₂O per acre); for high-yielding coarse varieties nitrogen requirement is higher, up to 120-140 kg N/ha. Full phosphorus and potassium and about one-third of nitrogen are incorporated at final puddling, with the remainder of nitrogen split at active tillering and panicle initiation stages. Zinc sulphate application (25 kg/ha) is important on zinc-deficient soils common to the rice belt, addressing "khaira disease" (zinc deficiency symptom in rice, characterized by bronzing/stunting of young seedlings).

3.8 Water Management After Transplanting

Standing water of 5-7 cm depth is generally maintained through active vegetative growth, with the field drained briefly at maximum tillering to encourage root development, then re-flooded through flowering and grain-filling, and finally drained about 10-15 days before harvest to facilitate combine harvesting and grain maturity.

3.9 Major Pests

- Stem borer (yellow stem borer, *Scirpophaga incertulas*): causes "dead heart" in vegetative stage and "white ear" at reproductive stage; managed via resistant varieties, light traps, and targeted insecticide application.
- Rice leaf folder: folds leaves and scrapes chlorophyll, reducing photosynthetic area; controlled with recommended insecticides at threshold levels.

- Brown planthopper (BPH) and whitebacked planthopper: sap-sucking pests that can cause "hopper burn" (rapid plant death) in dense, over-fertilized crops; managed by avoiding excess nitrogen, conserving natural predators, and targeted insecticide use.
- Rice hispa: leaf-feeding beetle, more localized outbreaks.

3.10 Major Diseases

- Rice blast (*Magnaporthe oryzae*): most serious fungal disease, affecting leaves, nodes, and panicles ("neck blast" causes severe yield loss); managed with resistant varieties, balanced nitrogen (avoiding excess), and fungicide (tricyclazole) where needed.
- Bacterial leaf blight (*Xanthomonas oryzae*): causes yellowing and wilting of leaves, especially serious in Basmati; managed with resistant varieties and avoiding field injury during management operations.
- Brown spot: associated with nutrient-deficient soils; addressed through balanced fertilization.
- Sheath blight: occurs in dense, high-nitrogen crops with poor aeration.

3.11 Harvesting and Yield

Basmati rice matures in 140-150 days (photoperiod-sensitive, so harvest timing is influenced by day length as well as sowing date), typically harvested October-November. Coarse varieties mature faster (110-130 days) and are harvested earlier. National average paddy yield is around 2.5-3.0 tonnes/ha for Basmati and higher (4-5 tonnes/ha) for coarse high-yielding varieties. Harvested paddy is dried to about 12-14 percent moisture before milling.

3.12 Milling and Post-Harvest

Paddy is milled to remove husk (producing brown rice) and then polished to produce white rice; milling recovery (the percentage of rice obtained from paddy) is a key quality and economic parameter, typically 60-68 percent depending on variety and processing quality. Basmati rice is often aged for several months to a year after milling, which improves cooking quality (aroma intensification and elongation ratio), commanding higher export prices for aged stock.

3.13 Economics and Export

Rice, especially Basmati, is one of Pakistan's top agricultural export commodities, with major destinations including the Middle East (particularly UAE, Saudi Arabia), Europe (UK), Kenya, and other African markets. The Rice Exporters Association of Pakistan (REAP) and the Trading Corporation of Pakistan play roles in export facilitation and quality standards enforcement (including pesticide residue compliance for EU markets, which has periodically caused trade friction).

Chapter 4: Maize (*Zea mays*) — Makai/Bhutta

4.1 Overview and Importance

Maize is Pakistan's third most important cereal crop after wheat and rice, and the fastest-growing in terms of yield improvement due to widespread adoption of hybrid seed. It serves triple purposes: human food (especially in KP where maize roti is a staple), poultry and livestock feed (the dominant modern use, driving demand growth), and industrial raw material (starch, corn oil, and increasingly ethanol). Area under maize is around 1.3-1.4 million hectares with production exceeding 8-9 million tonnes, reflecting some of the highest per-hectare yield gains of any major crop in Pakistan due to hybrid adoption.

4.2 Growing Regions and Seasons

Maize is grown in two seasons:

- Spring maize (irrigated, Punjab): sown January-February, harvested May-June; grown mainly in Punjab (Chiniot, Sahiwal, Kasur, Okara) as a high-yielding hybrid crop, often followed by rice or fodder in the same year.
- Autumn/Kharif maize: sown July-August, harvested October-November; grown across Punjab and KP.
- KP (Khyber Pakhtunkhwa) is the traditional maize heartland of Pakistan, particularly Swat, Mardan, Swabi, and Malakand, where it is a staple food crop grown by smallholders, often as open-pollinated varieties (OPVs) alongside hybrids.
- Punjab is now the leading maize-producing province by volume due to widespread hybrid adoption in irrigated spring maize.

4.3 Climate and Soil Requirements

Maize requires warm temperatures (optimum 21-30°C) and is sensitive to frost at all growth stages. It grows best on well-drained, fertile loam soils with pH 5.5-7.5 and does not tolerate waterlogging, which causes rapid root damage and yield loss (maize is far more sensitive to waterlogging than rice or even wheat). Good soil aeration and organic matter content are important for the crop's relatively shallow but vigorous root system.

4.4 Major Varieties/Hybrids Cultivated in Pakistan

- Hybrid maize (dominant in Punjab, spring season): multinational hybrids (Monsanto/Bayer's Dekalb series, Syngenta hybrids, Pioneer hybrids) alongside local hybrids developed by Maize and Millets Research Institute (MMRI) Yousafwala/Sahiwal.
- Open-pollinated varieties (OPVs, common in KP smallholder systems): Azam, Sarhad White, Sarhad Yellow, Pahari — lower yielding but cheaper as farmers can save seed, an important consideration for resource-poor hill farmers.

- Fall armyworm-tolerant and drought-tolerant lines are increasingly being promoted following the fall armyworm invasion.
- Baby corn and sweet corn varieties are grown in smaller peri-urban areas for fresh vegetable markets.

4.5 Land Preparation and Sowing

Fields require 2-3 ploughings and planking for a fine seedbed. Sowing is typically done in ridges/furrows for irrigated hybrid maize (allowing furrow irrigation and better drainage) at row spacing of 75 cm and plant-to-plant spacing of 18-20 cm, targeting a population of roughly 60,000-80,000 plants/ha for hybrids (higher densities, up to 100,000+/ha, are used for specialized high-density hybrids). Seed rate is approximately 20-25 kg/ha for hybrids (dibbled/drilled, precision planted where possible) and higher for OPVs.

4.6 Irrigation Requirements

Maize requires 5-8 irrigations over its growth cycle depending on season and rainfall, with critical stages being knee-high/vegetative stage, tasseling/silking (the most water-sensitive stage — moisture stress here severely reduces kernel set), and grain-filling. Spring maize, grown in the hot pre-monsoon period, requires more frequent irrigation than autumn maize which can partly rely on monsoon rains.

4.7 Fertilizer/Nutrient Management

Hybrid maize is a heavy feeder, requiring approximately 200-250 kg N, 100-125 kg P₂O₅, and 60-75 kg K₂O per hectare for high-yield production (roughly 80-100 kg N, 40-50 kg P₂O₅, 25-30 kg K₂O per acre), considerably higher than wheat. Phosphorus and potassium and about one-quarter of nitrogen are applied basally; remaining nitrogen is split into 2-3 top-dressings at knee-high, pre-tasseling, and silking stages. Zinc deficiency is also common in maize on calcareous soils and is corrected with zinc sulphate.

4.8 Major Pests

- Fall armyworm (*Spodoptera frugiperda*): an invasive pest first confirmed in Pakistan in 2019, now a major threat to maize nationwide; larvae feed inside the whorl, causing characteristic window-pane damage and frass; managed through early detection (pheromone traps), timely insecticide application (emamectin benzoate, chlorantraniliprole) targeted into the whorl, and encouraging natural enemies.
- Maize stem borer: bores into stems causing "dead heart" in young plants; managed with resistant hybrids and targeted insecticide.
- Termites: particularly in light, sandy soils.

4.9 Major Diseases

- Maize streak virus and other viral diseases: transmitted by leafhoppers, managed through vector control and resistant hybrids.

- Leaf blights (Turcicum/northern corn leaf blight, southern corn leaf blight): fungal diseases favored by humid conditions, managed via resistant hybrids and fungicide where severe.
- Stalk rots: associated with imbalanced fertilization (excess N, deficient K) and lodging risk; managed through balanced nutrition.
- Downy mildew: more localized, favored by high humidity at early growth stages.

4.10 Harvesting and Yield

Maize matures in 90-120 days depending on variety/hybrid and season. Harvest occurs when husks turn brown/dry and kernel moisture drops to around 20-25 percent for hand harvest (further dried afterward), or lower for direct combine harvesting. National average yield is around 4.5-5.0 tonnes/ha overall, but irrigated hybrid spring maize in Punjab regularly achieves 8-10 tonnes/ha or higher on progressive farms, among the highest cereal yields achieved in Pakistani agriculture — a key reason maize area has expanded rapidly as farmers substitute it for lower-return crops.

4.11 Post-Harvest and Storage

Proper drying to below 13-14 percent moisture is essential before storage to prevent aflatoxin contamination (a significant food safety and export quality concern for maize globally, caused by *Aspergillus flavus* fungal growth under humid storage conditions). Maize is stored in bags or, increasingly, in metal/plastic silos at commercial scale for the poultry feed industry, which is the largest domestic consumer of maize grain in Pakistan.

4.12 Economics and Use

The dominant demand driver for maize in Pakistan today is the poultry feed industry, followed by human consumption (particularly in KP) and a growing corn processing sector (starch, glucose, corn oil). Maize prices are more market-driven (less government price intervention) than wheat, making profitability closely tied to poultry sector demand cycles and input costs of hybrid seed, which farmers must purchase fresh each season (unlike OPVs or wheat/rice, hybrid seed cannot be effectively saved for replanting).

Chapter 5: Potato (*Solanum tuberosum*) — Aloo

5.1 Overview and Importance

Potato is Pakistan's leading vegetable crop by both area and economic value, and increasingly an export commodity to Central Asia, the Gulf, Sri Lanka, and Russia. Punjab dominates production, contributing over 85 percent of national output, with area around 190,000-220,000 hectares and total production typically 4-6 million tonnes, making Pakistan one of the larger potato producers in South Asia.

5.2 Growing Regions and Seasons

- Punjab (autumn/main crop): Okara, Sahiwal, Kasur, Pakpattan — sown September-October, harvested December-January; this is the largest crop by volume and the main source of seed potato for the spring crop.
- Punjab (spring crop): sown January-February, harvested April-May, using seed saved (cold-stored) from the autumn crop.
- KP (summer crop, high-altitude): Swat, Abbottabad, Mansehra — sown March-April, harvested August-September, benefiting from cool hill climate; this region is also an important source of certified seed potato due to lower virus pressure at altitude.
- Balochistan: Quetta, Pishin, Kalat — summer crop similarly benefiting from cool highland conditions.

5.3 Climate and Soil Requirements

Potato is a cool-season crop, performing best with day temperatures of 18-22°C and night temperatures of 12-15°C; tuber formation is inhibited above about 30°C, which limits Punjab's autumn/spring crops to cooler months while allowing KP and Balochistan highlands to grow it in summer. Potato requires well-drained, loose, sandy-loam soil with good organic matter content and pH 5.5-6.5; heavy clay or waterlogged soils cause poor tuber shape and increase disease incidence.

5.4 Major Varieties Cultivated in Pakistan

- Cardinal (red-skinned, widely grown, though older/losing ground to newer varieties), Desiree, Diamant, Lady Rosetta (processing/chips variety), Sante, Kuroda, FD (Faisalabad Desiree types), Lal Tor.
- Newer varieties promoted for yield and disease resistance include Caesar, Markies, and various imported Dutch and other European varieties adapted through trials by the Federal Seed Certification Department and provincial research stations.
- Seed potato is largely imported (from Netherlands and other sources) for basic/foundation seed multiplication, given Pakistan's limited domestic certified seed production capacity relative to demand, though KP hill areas are being developed as domestic seed potato production zones.

5.5 Land Preparation and Planting

Deep ploughing (2-3 times) followed by ridging is standard, since potatoes are planted on ridges/beds to facilitate tuber development and drainage. Seed tubers (30-40g, whole or cut with at least 2 eyes) are planted at 20-25 cm spacing on ridges 60-75 cm apart, giving a seed rate of roughly 1.5-2.5 tonnes/ha depending on seed size and spacing. Seed should be sprouted (chitted) before planting for more uniform emergence.

5.6 Irrigation Requirements

Potato requires frequent light irrigations (7-10 irrigations over the crop cycle) since its shallow root system is sensitive to both drought and waterlogging. Critical stages are tuber initiation (about 30-40 days after planting) and tuber bulking (40-70 days); moisture stress during these stages sharply reduces yield and tuber quality (causing hollow heart and secondary growth/knobbiness).

5.7 Fertilizer/Nutrient Management

Potato is a heavy feeder requiring approximately 150-180 kg N, 100-125 kg P₂O₅, and 125-150 kg K₂O per hectare — notably higher potassium requirement than cereal crops, reflecting the crop's need for potassium in tuber bulking and starch formation. Full P and K plus part of N is applied basally at planting/ridging, with remaining N applied at earthing-up (about 25-30 days after planting, when ridges are built up around growing plants to prevent tuber greening from light exposure).

5.8 Major Pests

- Potato tuber moth (*Phthorimaea operculella*): larvae bore into tubers both in field and storage, a major post-harvest loss driver; managed via proper earthing-up (preventing tuber exposure), field sanitation, and appropriate storage practices.
- Aphids: primary concern as virus vectors (transmitting potato leafroll virus and potato virus Y, which degrade seed quality over successive generations) rather than direct feeding damage; managed via insecticide sprays and use of certified virus-free seed.
- Cutworms: damage young seedlings at soil level.

5.9 Major Diseases

- Late blight (*Phytophthora infestans*): the most destructive potato disease worldwide and in Pakistan, especially damaging in humid, cool weather; causes rapid foliage collapse and tuber rot; managed through resistant/tolerant varieties, protective fungicide sprays (mancozeb, metalaxyl-based systemic fungicides), and destroying infected haulms before harvest.
- Early blight (*Alternaria solani*): less severe than late blight but widespread; managed with fungicide sprays.
- Bacterial wilt and blackleg: soil and seed-borne bacterial diseases managed through certified disease-free seed and crop rotation.

- Common scab: affects tuber skin appearance/marketability, associated with alkaline soil and dry conditions during tuber formation.

5.10 Harvesting, Yield, and Storage

Potato matures in 90-120 days depending on variety and season. Haulms (foliage) are typically cut/killed about 10-15 days before harvest to allow skin-set, which improves storage life. National average yield is around 20-22 tonnes/ha, with progressive Punjab farms achieving 30-35 tonnes/ha under good management. Cold storage (2-4°C, high humidity) is critical infrastructure, particularly for the autumn crop, which must be stored for months to supply seed for the spring crop and fresh market demand through the summer; Pakistan has significant cold storage capacity concentrated in Punjab (Okara/Sahiwal belt), though storage capacity gaps and post-harvest losses remain a persistent challenge, especially in bumper-crop years that depress farm-gate prices.

5.11 Economics and Export

Potato is a significant export crop for Pakistan, with major destinations including Afghanistan, Sri Lanka, UAE, Russia, and Central Asian states (Uzbekistan, Kazakhstan, Tajikistan). Domestic price volatility is high due to the crop's relatively short storage-adjusted market window and boom-bust planting cycles (farmers overplanting after a profitable year, leading to gluts and price crashes the following season).

Chapter 6: Tomato (*Solanum lycopersicum*) — Tamatar

6.1 Overview and Importance

Tomato is one of Pakistan's most widely consumed vegetables and is notorious for extreme price volatility, occasionally becoming a major inflation and political talking point when supply shortfalls (often due to weather damage to the Sindh/Balochistan winter crop or cross-border trade disruptions with Iran/Afghanistan) send retail prices sharply higher. Area under tomato is roughly 60,000-65,000 hectares with production around 550,000-600,000 tonnes, though domestic production still falls short of demand in certain seasons, historically requiring imports.

6.2 Growing Regions and Seasons

- Sindh (Umerkot, Mirpurkhas, Hyderabad): major winter-season production supplying the country during months when Punjab cannot grow tomato outdoors.
- Balochistan (Mastung, Kalat, Pishin): summer production benefiting from cool highland climate, supplying markets when lowland production is out of season.
- Punjab (multiple zones): both spring and autumn crops depending on district, plus significant tunnel/greenhouse cultivation near urban centers for off-season and premium markets.
- KP: smaller-scale production in various districts, including hill areas for summer crop.

6.3 Climate and Soil Requirements

Tomato grows best at temperatures of 20-27°C; fruit set is impaired above 35°C or below 10°C, and flowers may drop under heat stress, which is a major cause of Pakistan's summer tomato price spikes (lowland Punjab and Sindh become too hot for good fruit set in peak summer). It requires well-drained sandy-loam to loam soil with pH 6.0-7.0 and good organic matter; waterlogging causes root rot and poor fruit quality.

6.4 Major Varieties Cultivated in Pakistan

- Open-field varieties: Roma, Rio Grande, Money Maker, Nagina, Sahil.
- Hybrid varieties (increasingly dominant, especially in tunnel farming): various F1 hybrids from multinational and local seed companies bred for higher yield, disease tolerance (particularly Tomato Yellow Leaf Curl Virus tolerance), and longer shelf life for transport.
- Tunnel/greenhouse cultivation using hybrid indeterminate varieties is expanding around Lahore, Islamabad, and other urban centers to enable off-season, higher-value production.

6.5 Nursery and Transplanting

Tomato is typically raised in a nursery for 25-30 days before transplanting into the main field, at spacing of about 60 x 45 cm (row x plant), though tunnel cultivation with staking/trellising uses closer spacing with vertical training. Seed rate is approximately 300-400 g/ha for nursery raising.

6.6 Irrigation and Fertilizer Requirements

Tomato requires frequent, moderate irrigation (drip irrigation increasingly used in commercial and tunnel production for water efficiency and disease reduction, since overhead/flood irrigation increases foliar disease risk); waterlogging and moisture stress both cause blossom-end rot and fruit cracking. Recommended fertilizer dose is approximately 120-150 kg N, 80-100 kg P₂O₅, and 100-120 kg K₂O per hectare, with potassium especially important for fruit quality; calcium deficiency (often linked to irregular watering rather than true soil deficiency) causes blossom-end rot, a very common quality problem reported by growers.

6.7 Major Pests

- Whitefly (*Bemisia tabaci*): the most serious pest, both for direct sap-feeding damage and as the vector of Tomato Yellow Leaf Curl Virus (TYLCV), which can devastate crops, especially in Sindh and Punjab during warm months; managed via yellow sticky traps, insecticide rotation, virus-tolerant varieties, and insect-proof netting in tunnel systems.
- Fruit borer (*Helicoverpa armigera*): bores into developing fruit causing direct marketable yield loss; managed with pheromone traps and targeted insecticide application.
- Leaf miner and thrips: cause foliar damage and can also vector viruses.

6.8 Major Diseases

- Early blight and late blight (*Alternaria* and *Phytophthora*, same late blight pathogen affecting potato): foliar and fruit diseases favored by humid conditions; managed with fungicide sprays and crop rotation away from potato/other solanaceous crops.
- Bacterial wilt: soil-borne, causes rapid wilting; managed through resistant rootstock (in some grafted tunnel systems) and crop rotation.
- Tomato Yellow Leaf Curl Virus (TYLCV): whitefly-transmitted, causes severe stunting and yield loss; the single most damaging disease in many production seasons, managed primarily through vector (whitefly) control and resistant varieties.

6.9 Harvesting, Yield, and Marketing

Tomato reaches first harvest about 70-90 days after transplanting and is picked over multiple rounds (indeterminate varieties/tunnel crops especially can be harvested over an extended period). National average yield is around 9-10 tonnes/ha, though tunnel/hybrid commercial production can exceed 40-50 tonnes/ha under intensive management. Because tomato is highly perishable and lacks significant cold-chain infrastructure across most of the supply chain, post-harvest losses are substantial, and price crashes at the farm gate during gluts (followed by shortages and price spikes weeks later) are a chronic feature of the market, driven partly by weather-related supply shocks in the Sindh/Balochistan production windows.

Chapter 7: Black Pepper (*Piper nigrum*) — Kali Mirch

7.1 Overview and Pakistan Context

Unlike maize, wheat, rice, potato, and tomato, black pepper is not commercially cultivated in Pakistan at meaningful scale — the crop requires a humid tropical climate with consistent rainfall (typically found in regions like India's Kerala/Karnataka, Vietnam, Indonesia, and Brazil) that does not exist in Pakistan's predominantly arid and semi-arid agro-ecological zones. Pakistan is therefore entirely import-dependent for black pepper, sourcing primarily from Vietnam, India, Sri Lanka, and Indonesia through commercial spice importers based in Karachi. This chapter is included for RAG completeness/reference purposes given the crop's importance in Pakistani cuisine, and to correctly advise any agricultural advisory system that black pepper is not a viable commercial recommendation for Pakistani farmers under current climatic conditions, with the limited exception of small experimental or hobbyist cultivation in very high-humidity microclimates (parts of the Kashmir foothills or greenhouse settings) that has no commercial scale.

7.2 Botanical and Agronomic Requirements (General Reference)

Black pepper is a perennial woody vine that requires:

- Consistently high humidity (70-90 percent) and well-distributed annual rainfall of 2,000-3,000 mm.
- Warm, stable temperatures of 23-32°C without significant frost or extended dry periods.
- Support structures (living or dead standing trees/poles) for the climbing vine.
- Rich, well-drained lateritic or forest soils high in organic matter.

These conditions are fundamentally incompatible with Pakistan's climate outside of small artificial (greenhouse/shade-house) environments, which would not be commercially viable at field-crop scale.

7.3 Role in Pakistani Agriculture and Trade

Black pepper functions in Pakistan's agricultural economy purely as an imported spice commodity, processed (cleaned, sometimes ground) domestically by the spice trade and food processing sector centered in Karachi, and used extensively in Pakistani cuisine (garam masala blends, meat dishes, biryani). For a Pakistan-focused agricultural advisory or RAG system, queries about black pepper cultivation should generally redirect users toward crops with genuine climatic suitability in their region, while any query about black pepper sourcing, pricing, or culinary use can be addressed as an imported commodity topic rather than a domestic cultivation recommendation. Farmers interested in high-value spice cultivation suited to Pakistan's actual climate are better directed toward crops such as chili, turmeric, ginger, coriander, and cumin, which are covered elsewhere in this reference and are genuinely grown domestically.

Chapter 8: Apple (*Malus domestica*) — Seb

8.1 Overview and Importance

Apple is Pakistan's leading temperate fruit crop by both area and value, concentrated almost entirely in the high-altitude mountainous regions of the north where winter chilling requirements can be met. It is a major cash crop and livelihood source for farmers in Balochistan and Gilgit-Baltistan, and total national production is typically in the range of 400,000-600,000 tonnes annually, though yields and quality lag behind leading international producers due to aging orchards, limited use of modern dwarfing rootstocks, and post-harvest infrastructure gaps.

8.2 Growing Regions

- Balochistan: the largest apple-producing province, particularly Quetta, Pishin, Qila Saifullah, Ziarat, Loralai — high-altitude, dry temperate climate ideal for quality apple production with distinct chilling hours in winter.
- Gilgit-Baltistan: Skardu, Hunza, Gilgit — traditional apple growing areas with excellent quality but more limited market access due to remoteness and seasonal road closures.
- Khyber Pakhtunkhwa: Swat, Chitral, Kalam, and other hill areas — smaller-scale but expanding cultivation.
- Azad Jammu & Kashmir: some cultivation in higher-elevation areas.

8.3 Climate and Chilling Requirements

Apple requires a defined winter dormancy period with adequate chilling hours (generally 800-1,600 hours below 7°C depending on variety) to break dormancy properly and ensure uniform flowering the following spring; insufficient chilling causes erratic, prolonged bloom and reduced fruit set — an increasing concern as winters warm. Apple trees need well-drained soils, cold winters, and moderately warm (not excessively hot) summers, which is why cultivation is restricted to elevations generally above 1,800-2,500 meters in Pakistan's context, distinguishing it sharply from the lowland irrigated-plain crops covered elsewhere in this reference.

8.4 Major Varieties Cultivated in Pakistan

- Red/Royal Delicious and various Delicious strains: historically dominant, though quality (color development, storage life) is often inferior to imported apples due to orchard management practices, prompting consumer preference shifts toward imported fruit in urban premium markets.
- Kala Kulu, Amri (Ambri), Mashadi: traditional local varieties, especially significant in Balochistan and Gilgit-Baltistan, valued for local adaptation though generally lower yielding and less uniform than modern cultivars.
- Golden Delicious, Gala, Fuji: increasingly planted in newer orchards for better market quality and price, though total area remains limited relative to older Delicious-type plantings.

- Efforts by research institutes and provincial horticulture departments have promoted dwarfing rootstocks (M9, M26, MM106) and high-density orchard systems to improve yield per hectare and earlier bearing, replacing the traditional seedling rootstock orchards that dominate older plantings and are slow to come into production (7-10 years) with wide, inefficient spacing.

8.5 Orchard Establishment

Traditional Pakistani apple orchards use seedling rootstocks with wide spacing (typically 6 x 6 m or wider), resulting in large but slow-maturing, low-density orchards (roughly 250-275 trees/ha) that take 6-8 years to reach significant bearing and complicate spraying/harvesting operations. Modern high-density systems using dwarfing rootstocks can achieve 800-2,500+ trees/ha with earlier bearing (2-3 years) and higher yield per hectare, though adoption remains limited due to higher establishment cost, need for irrigation/trellising infrastructure, and limited nursery availability of certified dwarfing rootstock planting material domestically.

8.6 Irrigation and Fertilizer Management

Apple orchards in Balochistan are predominantly irrigated via tubewells (groundwater), and the region faces a serious and worsening groundwater depletion crisis due to over-extraction for orchard and general agricultural irrigation combined with low rainfall recharge — a critical sustainability concern for the sector's future. Fertilizer application is typically farmyard manure plus NPK, applied in early spring before bud break and sometimes split with a post-harvest application, though many older orchards receive inconsistent or inadequate fertilization, contributing to declining productivity and fruit quality over tree lifespan.

8.7 Pruning and Training

Annual dormant-season pruning (winter, after leaf fall) is essential for maintaining tree structure, light penetration, and fruiting wood renewal; open-center or modified central leader training systems are used, though many traditional orchards suffer from inadequate or improper pruning leading to overcrowded canopies, poor fruit color development, and increased disease pressure due to poor air circulation.

8.8 Major Pests

- Codling moth (*Cydia pomonella*): the most economically damaging apple pest globally and in Pakistan, larvae tunnel into fruit rendering it unmarketable; managed through pheromone mating disruption, targeted insecticide sprays timed to egg hatch, and orchard sanitation (removing fallen/infested fruit).
- San Jose scale: sap-sucking scale insect that weakens trees and blemishes fruit; managed with dormant-season oil sprays.
- Woolly apple aphid: causes gall formation on roots and branches; managed with targeted insecticides and biological control (parasitic wasps where established).
- Fruit fly: increasingly problematic in some regions, managed with bait traps and sanitation.

8.9 Major Diseases

- Apple scab (*Venturia inaequalis*): fungal disease causing dark lesions on leaves and fruit, favored by humid spring conditions; managed with protectant fungicide sprays during the critical infection period around bloom.
- Fire blight (*Erwinia amylovora*, bacterial): causes shoot dieback with a characteristic "shepherd's crook" wilting symptom and can be highly destructive when present; managed through sanitation (pruning out infected wood well below visible symptoms, disinfecting tools) and, where necessary, bactericide sprays during bloom.
- Powdery mildew: affects young shoots and leaves in humid conditions; managed with sulfur-based fungicides.
- Root rots (*Armillaria* and *Phytophthora* species): associated with poor drainage and old orchard sites; managed through site selection and drainage improvement in new plantings.

8.10 Harvesting, Yield, and Post-Harvest

Harvest timing depends on variety and elevation, generally August-October, determined by fruit firmness, background color change, and starch-iodine testing in more advanced operations (though many smallholder orchards rely on visual/experience-based judgment alone). National average yield remains comparatively low (often cited around 7-10 tonnes/ha) relative to major international apple-producing countries, reflecting the prevalence of old seedling-rootstock orchards, inconsistent management, and post-harvest handling losses. Cold storage capacity in Balochistan and Gilgit-Baltistan is limited relative to production, so a large share of the harvest must be marketed within weeks of picking, depressing farm-gate prices during the harvest glut and contributing to significant post-harvest losses from bruising, poor grading, and inadequate packaging during transport to distant urban markets (particularly Karachi and Lahore).

8.11 Economics and Marketing Challenges

The apple value chain in Pakistan faces structural challenges including fragmented, small orchard holdings, limited access to certified planting material of modern varieties/rootstocks, weak cold-chain and grading infrastructure, and competition from imported apples (from countries like Iran, China, and others) in urban premium retail segments where consumers often perceive imported fruit as higher quality. Government and donor-funded horticulture development programs have periodically supported orchard rehabilitation, drip irrigation adoption, and high-density plantation demonstrations in Balochistan and GB to address these structural issues.

Chapter 9: Cotton (*Gossypium hirsutum*) — Kapas

9.1 Overview and Importance

Cotton is historically Pakistan's most economically significant cash crop, feeding the country's large textile and garment export industry, which has long been one of the largest sources of foreign exchange earnings for the national economy. Area under cotton is typically 2.0-2.5 million hectares, concentrated heavily in Punjab and Sindh, with production measured in millions of bales (a bale being a standard packed unit of ginned cotton fiber).

9.2 Growing Regions

- Punjab: Multan, Bahawalpur, Vehari, Rahim Yar Khan, Dera Ghazi Khan, Muzaffargarh — the traditional cotton belt of southern Punjab.
- Sindh: Sanghar, Nawabshah, Mirpurkhas, Hyderabad — the second major cotton-producing zone.
- Balochistan: smaller-scale cultivation in irrigated pockets.

9.3 Climate and Soil Requirements

Cotton requires a long, warm frost-free growing season of at least 180-200 days, with temperatures of 25-35°C during the main growth period; it is highly sensitive to cold temperatures, especially at germination and boll development. Cotton performs best on well-drained, deep loam to clay-loam soils with good water-holding capacity and pH 6.0-8.0, tolerating moderate salinity better than many other row crops, which is one reason it has historically been favored in parts of Sindh and southern Punjab affected by soil salinity.

9.4 Major Varieties Cultivated in Pakistan

Bt (genetically modified, insect-resistant) cotton hybrids now dominate Pakistani cotton cultivation, having been widely adopted (though often through informal/unregistered seed channels in earlier years) for their resistance to bollworm pests. Popular varieties/hybrids include IUB-13, FH-142, CIM-599, CIM-602, and various commercial Bt hybrids developed by both public research institutes (Central Cotton Research Institute Multan, Cotton Research Station Sakrand) and private seed companies.

9.5 Land Preparation and Sowing

Cotton is typically sown from mid-April to May in Punjab (slightly later in Sindh), following 2-3 ploughings and planking. Sowing is done in ridges/furrows at row spacing of 75 cm and plant-to-plant spacing of 22-30 cm, with seed rate of approximately 12-15 kg/ha for delinted seed.

9.6 Irrigation and Fertilizer

Cotton requires 6-8 irrigations over its extended growing season, with critical stages being squaring (bud formation), flowering, and boll development. Fertilizer recommendation is approximately 100-150 kg N, 60-90 kg P₂O₅, and 50-60 kg K₂O per hectare, with nitrogen split across sowing, squaring,

flowering, and boll development stages due to the crop's extended fruiting period. Boron deficiency, which causes flower and boll shedding, is a recognized concern managed with foliar boron sprays during flowering.

9.7 Major Pests

- Whitefly (*Bemisia tabaci*): the primary vector of Cotton Leaf Curl Virus (CLCuV), one of the most economically damaging cotton diseases in Pakistan; also causes direct sap-feeding damage and sooty mold growth on honeydew secretions.
- Pink bollworm and American bollworm: bore into cotton bolls, causing direct fiber and seed damage; historically the primary driver of Bt cotton adoption.
- Jassids (leafhoppers): cause characteristic leaf reddening and curling ("hopper burn").
- Mealybug: a more recently emerged pest causing stunting and honeydew-related sooty mold.

9.8 Major Diseases

- Cotton Leaf Curl Virus (CLCuV): whitefly-transmitted, causes severe leaf curling, vein thickening, and stunting; historically one of the most damaging cotton diseases in Pakistan's history, having caused major production crises in the 1990s; managed primarily through resistant/tolerant varieties and whitefly vector control.
- Bacterial blight: causes angular leaf spots and boll rot in humid conditions.

9.9 Harvesting, Yield, and Economics

Cotton is harvested by hand-picking over multiple rounds (2-4 pickings) as bolls mature and open, typically September through November. National average yield is around 650-750 kg lint per hectare, though this has been volatile in recent years due to pest pressure, climate stress, and input cost challenges. Ginning (separating fiber from seed) is done at ginning factories concentrated in the cotton belt, with cottonseed itself being a valuable by-product used for cottonseed oil and cottonseed cake (livestock feed).

Chapter 10: Sugarcane (*Saccharum officinarum*) — Ganna

10.1 Overview and Importance

Sugarcane is Pakistan's major sugar-industry feedstock crop and one of the few crops occupying the field for an extended 10-12 month (or longer) duration. Area under sugarcane is typically 1.0-1.3 million hectares, concentrated in Punjab and Sindh, feeding the country's large sugar milling industry.

10.2 Growing Regions and Seasons

- Punjab: Faisalabad, Jhang, Toba Tek Singh, Multan, Rahim Yar Khan — spring planting (February-March) is dominant.
- Sindh: Mirpurkhas, Hyderabad, Tando Allahyar — both spring and autumn (September-October) planting are practiced.

10.3 Climate and Soil Requirements

Sugarcane requires a long, warm growing season with temperatures of 25-35°C and high water availability throughout its extended growth period; it performs best on deep, well-drained loam soils with good moisture-retention capacity, since it is one of the most water-intensive crops grown in Pakistan.

10.4 Fertilizer and Irrigation

Sugarcane requires approximately 200-250 kg N, 100-125 kg P₂O₅, and 100-125 kg K₂O per hectare, with phosphorus and potassium applied basally and nitrogen split across multiple applications during the extended tillering and grand growth phases. Irrigation requirement is substantial, with 15-20+ irrigations typically needed over the full crop cycle depending on rainfall and season.

10.5 Major Pests and Diseases

- Sugarcane borers (top borer, stem borer): bore into the cane stalk, reducing sugar recovery and cane weight; managed through resistant varieties and biological control (parasitic wasps) alongside targeted insecticide use.
- Red rot: a serious fungal disease causing internal reddening and rotting of the cane, spread through infected seed cane; managed primarily through use of disease-free seed cane and resistant varieties, since chemical control of an internal stalk disease is difficult.

10.6 Harvesting and Economics

Sugarcane is harvested 10-12 months after planting (or earlier for ratoon crops, where the cane regrows from the previous crop's root system rather than being replanted), typically November through April, timed to supply sugar mills during their crushing season. National average yield is around 55-65 tonnes/ha. Cane is sold to sugar mills at a government-notified support price, and by-products include molasses (used in ethanol production and animal feed) and bagasse (fibrous residue, used as fuel for mill boilers or in paper/board manufacturing).

Chapter 11: Chickpea/Gram (*Cicer arietinum*) — Channa

11.1 Overview and Importance

Chickpea, locally known as chana or gram, is Pakistan's most important pulse (legume) crop and a key source of plant protein, particularly significant in rain-fed farming systems where few other crops are viable. Area under chickpea is typically around 900,000 hectares to 1 million hectares, with the majority grown under rain-fed conditions, meaning national production is closely tied to the timing and amount of monsoon and winter rainfall each year.

11.2 Growing Regions

- Punjab: rain-fed tracts of the Thal desert and Potohar Plateau (Chakwal, Jhelum, Attock) are the dominant chickpea zones.
- Sindh and KP: smaller-scale cultivation, both rain-fed and irrigated depending on district.

11.3 Climate and Soil Requirements

Chickpea is a cool-season crop growing well at average temperatures of 15-25°C, and it can tolerate significant drought stress once established, making it well-suited to Pakistan's rain-fed marginal lands where wheat or other cereals would underperform. It grows on a wide range of soil types but performs best on well-drained loam soils; waterlogging causes root rot and severe stand loss.

11.4 Major Varieties

Kabuli type (large, cream-colored seed, higher export value): CM 2008 and similar varieties tested by AARI and the Pulses Research Institute. Desi type (smaller, darker seed, primarily for domestic consumption): various locally adapted varieties bred for drought tolerance and blight resistance.

11.5 Fertilizer Requirement

As a nitrogen-fixing legume (through root nodule *Rhizobium* bacteria), chickpea requires only a small starter nitrogen dose of 15-20 kg/ha, with a higher phosphorus requirement of 60-75 kg/ha to support root and nodule development. Seed inoculation with *Rhizobium* culture is recommended on fields without recent chickpea history.

11.6 Major Diseases and Pests

- Ascochyta blight (fungal): causes lesions on leaves, stems, and pods, particularly damaging in cool, wet conditions; managed through resistant varieties and fungicide seed treatment.
- Fusarium wilt: soil-borne, causes sudden wilting; managed through resistant varieties and crop rotation.
- Pod borer (*Helicoverpa armigera*): the major insect pest, feeding directly on developing pods; managed with pheromone traps and targeted insecticide application at threshold levels.

11.7 Harvesting and Yield

Chickpea matures in approximately 120-150 days depending on variety and rainfall conditions, harvested March-April. National average yield is relatively low compared to irrigated crops, typically 700-900 kg/ha, reflecting its predominantly rain-fed, marginal-land cultivation; irrigated chickpea can achieve significantly higher yields where water is available.

Chapter 12: Onion (*Allium cepa*) — Pyaz

12.1 Overview and Importance

Onion is one of Pakistan's most widely consumed vegetable crops and a significant export commodity to the Middle East, particularly during periods of favorable regional price differentials. Area under onion is approximately 150,000-170,000 hectares, with Sindh and Balochistan being the dominant production zones.

12.2 Growing Regions and Seasons

- Sindh (Hyderabad, Mirpurkhas, Shaheed Benazirabad): the largest onion-producing region, primarily a winter/spring crop.
- Balochistan (Mastung, Kalat, Quetta): summer crop benefiting from cool highland conditions.
- Punjab: smaller-scale cultivation supplementing national supply.

12.3 Climate and Soil Requirements

Onion requires a cool growing period for vegetative growth followed by warmer, longer days for bulb formation; it grows best on well-drained sandy-loam soils with good organic matter and pH 6.0-7.5, and is sensitive to waterlogging, which causes bulb rot.

12.4 Fertilizer Requirement

Onion requires approximately 100-120 kg N, 60-80 kg P₂O₅, and 60-80 kg K₂O per hectare. Phosphorus and potassium are applied basally, while nitrogen is split into 2-3 applications, with the final nitrogen application timed before bulb initiation, since late nitrogen application after bulb formation begins can delay maturity and reduce storage quality.

12.5 Major Pests and Diseases

- Thrips: the primary insect pest, causing silvery leaf streaking and reduced bulb size under heavy infestation; managed with targeted insecticide sprays and blue sticky traps.
- Purple blotch (*Alternaria porri*): fungal disease causing purplish lesions on leaves, favored by humid conditions; managed with fungicide sprays.
- Basal rot and neck rot: post-harvest storage diseases, managed through proper curing (drying) of bulbs before storage and avoiding mechanical damage during harvest and handling.

12.6 Harvesting, Storage, and Economics

Onion matures in approximately 120-150 days depending on variety and season, with maturity indicated by neck-fall (foliage bending over) and drying. Proper curing (field-drying for several days after pulling) is essential for storage life. National average yield is around 12-15 tonnes/ha. Onion is a significant export crop for Pakistan to Gulf and Middle Eastern markets, though export volumes and

profitability fluctuate significantly based on regional supply conditions, particularly relative to India's onion production, which strongly influences South Asian regional price dynamics.